

ASHUTOSH SHARMA

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Generative AI Engineer

PROFESSIONAL SUMMARY

AI Engineer with 3.5+ years building production RAG and agentic systems for enterprise content automation and document intelligence. Delivered 5 end-to-end AI products, reducing operational costs by 50% across 3 major deployments.

TECHNICAL SKILLS

Generative AI & LLM: LlamaIndex, LangChain, RAG (Hybrid Search, CRAG), Agentic AI, ReAct Agents, Function Calling, Structured Output, Prompt Engineering, Guardrails, RAGAS, NLI Citation Verification, Hugging Face Transformers.

Backend & Infrastructure: Python (async), FastAPI, PostgreSQL, pgvector, SQL, Celery, RabbitMQ, WebSocket, REST APIs, Pydantic.

Cloud & DevOps: Docker, Kubernetes, Helm, Git, CI/CD, AWS S3, Object Storage, Phoenix Tracing, Prometheus.

ML & Data Science: PyTorch, scikit-learn, NLP, Deep Learning, Classification, Clustering, Topic Modeling, Feature Engineering, Pandas.

PROFESSIONAL EXPERIENCE

Data Scientist (Generative AI Engineering) | T-Systems ICT India Pvt. Ltd.

December 2022 - Present

AI Operator - Agentic RAG Platform | *May 2025 - Present*

- Design and build a multi-node RAG platform on LlamaIndex and FastAPI, lowering p95 response time by 35% under production load.
- Engineered ReAct agent across six retrieval tools, incorporating three new loop-prevention safeguards. Raised answer faithfulness from 40% to over 80% through NLI verification.
- Collaborate with cross-functional teams to extend ingestion from PDF-only to 6 formats onboarding 15+ data sources.
- Integrated RAGAS evaluation pipeline with Phoenix tracing for LLM observability. Streamlined prompt optimization across 1,000+ weekly evaluations and presented findings to VP of Engineering.

Insurance Healthcare Agent - Agentic RAG System | *December 2024 - April 2025*

- Developed agentic RAG system integrating document and PostgreSQL sources, improving query resolution efficiency by 40%.
- Partnered with domain experts to optimize agent workflows using function calling. Reduced agent workload by 50% and cut wait times by 70%.
- Spearheaded deployment processing 10,000+ monthly queries at 98% accuracy with resilient fallback handling and 99.5% uptime.

Crew AI Multi-Agent Content System | *July 2024 - November 2024*

- Coordinated multi-agent content pipeline using CrewAI with specialized writer, researcher, and editor agents across five distinct use cases.
- Accelerated content production by 80% on the blog pipeline while maintaining editorial quality standards.

Topic Modeling for Document Clustering | *February 2024 - July 2024*

- Analyzed 1,000+ enterprise documents using LDA topic modeling, boosting search and content organization by 45%.
- Trained a LightGBM classifier reaching 90% accuracy on extracted topics, eliminating 70% of manual categorization effort.

Enterprise Data Analytics & ML Modeling | December 2022 - January 2024

- Directed customer segmentation and classification models using XGBoost and Random Forest, reaching 87% prediction accuracy.
- Streamlined ETL pipelines processing 50K+ records with feature engineering, compressing data preparation time by 60%.
- Delivered interactive analytics dashboards and reporting pipelines for enterprise clients, improving decision cycles by 30%.

Intern Data Scientist | Analytics Labs

November 2021 - November 2022

- Constructed a Random Forest model for customer attrition analysis, driving a 12% reduction in targeted segment attrition.
- Established KPIs in the Memory Evaluation Matrix, enabling real-time CPE tracking across 500+ devices.

EDUCATION

Master of Science in Data Science | Manipal Academy of Higher Education, Manipal, India | July 2022 - July 2024

Bachelor of Science in Computer Science, Mathematics & Statistics | Bangalore North University, Bangalore, India | July 2018 - November 2021

CERTIFICATIONS

- Generative AI with Large Language Models - DeepLearning.AI / Coursera
- LangChain for LLM Application Development - DeepLearning.AI
- Machine Learning Specialization - Coursera (Andrew Ng)

LANGUAGES

English - Professional Working Proficiency | Hindi – Native